

ADR 0044 — Which host a fetch may contact: a name-shaped newtype *and* a precondition that refuses instead of being skipped

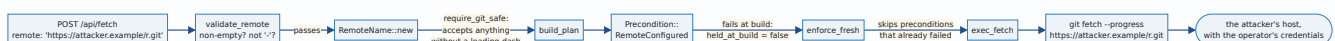
- **Status:** Accepted — implemented and tested.
- **Date:** 2026-08-02.
- **Milestone / issue:** M2.20c, issue #229 ("Fetch: `POST /api/fetch` with streamed progress and cancellation"), child of #73 (M2.20, "Remote operations"). Branch `feature/m2.20c-fetch-execution`, on top of ADR 0043.
- **Supersedes / superseded by:** Nothing superseded. **Corrects** a claim made by [0043](#) and by `docs/SECURITY_MODEL.md`'s "Remote and Forge Credentials" section: that `Precondition::RemoteConfigured` was what stopped a request from choosing the host this server connects to. It was not.
- **Related:** [0039](#) (where `FetchRemote`'s `RemoteConfigured` precondition was introduced), [0043](#) (fetch execution — the slice that made the gap reachable), [0002](#) (a request may not name a repository *path*, for the same reason it may not name a remote *URL*), [0028](#) (the sandbox constrains ports, explicitly **not** hosts — which is why this boundary has to hold in the application layer), [0042](#) §3 (the build/submit seam whose `held_at_build` census this ADR changes the meaning of), [0036](#) (the credential-bearing spawn this decides the destination of).

Context

ADR 0043 wired `POST /api/fetch` to a real `git fetch`. That made this server's first code path that opens a socket with the operator's credentials on it — a credential helper, an SSH agent, `~/.ssh/known_hosts` — reachable from an authenticated HTTP request. The handler's own module doc stated the defence:

`FetchRequest` carries a name, which the plan's `Precondition::RemoteConfigured` then requires to exist in the repository's own configuration. A request that could carry a URL would let any authenticated client point this server — and whatever credential helper or SSH agent the host offers it — at a host of the client's choosing.

The reasoning is right. The implementation did not do it. **Neither half held.**



Half one: the type accepted a URL

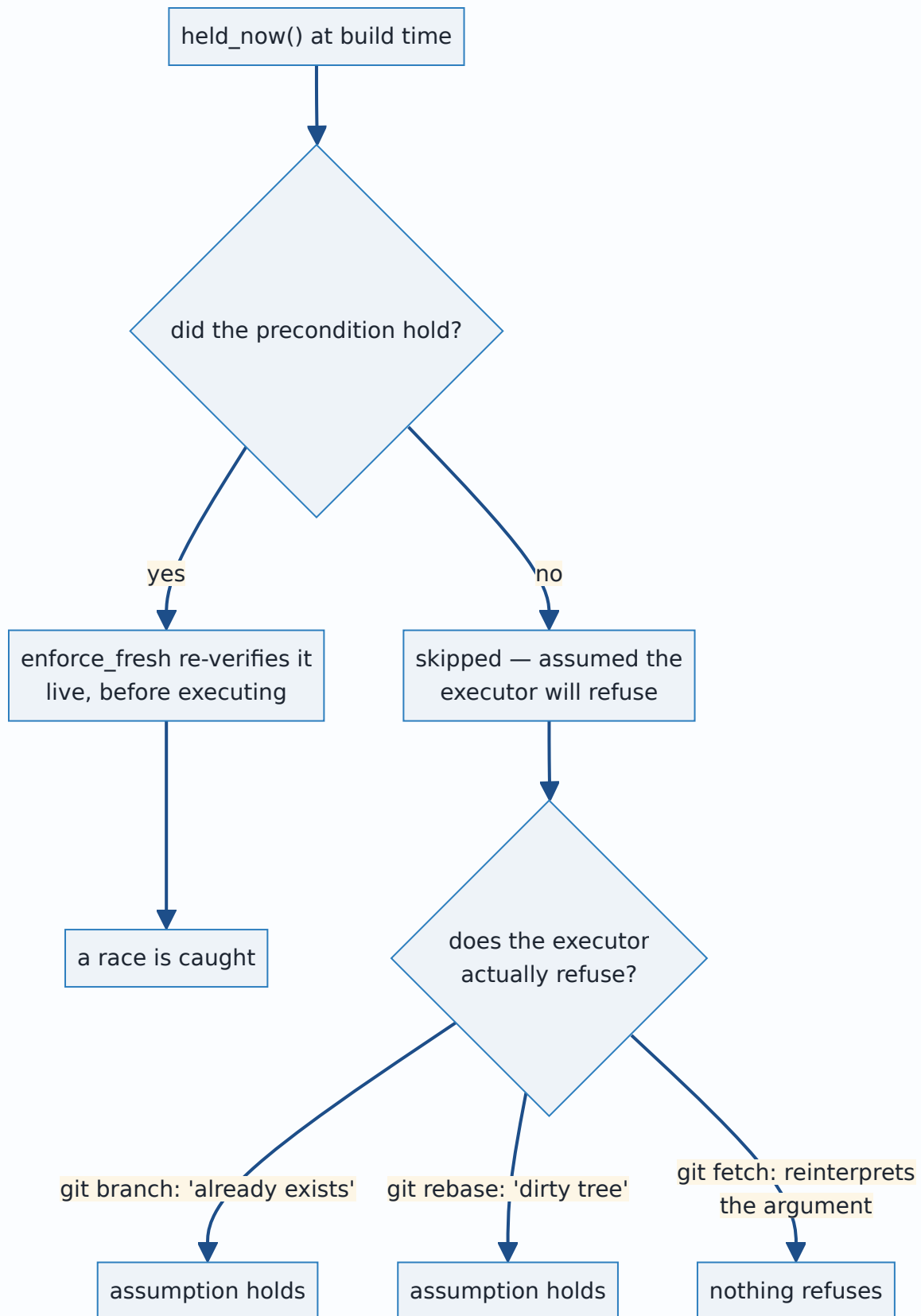
`RemoteName` was declared with `require_git_safe` — "non-empty, and does not start with -". `https://attacker.example/r.git` satisfies both. So the "carries a name, not a URL" sentence was a description of intent, not a property of the type. Nothing downstream re-checked the shape.

Half two: the precondition could not fire

`enforce_fresh` re-verifies **only the preconditions that held when the plan was built**. That is deliberate and, for every other precondition, correct — its own doc explains why:

One that already failed at build time is skipped here — the executor's own legacy guard refuses it with the exact wording it always had.

An unconfigured remote fails at build time. So the one precondition that was supposed to stop this was skipped *precisely in the case it existed for*.



Why git makes this worse than an ordinary missing check

`git fetch <arg>` does not reject an argument it cannot find among the configured remotes. It **falls through to treating it as a transport target**. Verified directly against git 2.43.0 on this host, before either fix existed:

```
$ git -C repo fetch ghost.git      # no 'ghost.git' remote is configured
From ghost
 * branch          HEAD          -> FETCH_HEAD
$ cat repo/.git/FETCH_HEAD
2e32d646837ebca218d9839ad717e6d2d40500a9      ghost
```

So a *well-formed* name that simply is not configured is silently resolved as a path relative to the worktree, and a URL-shaped one is resolved as a URL.

There is a second, quieter failure stacked on the first. `exec_fetch` reports what a fetch did by diffing `refs/remotes/<remote>/*` before and after (ADR 0037's posture). An ad-hoc target moves **no** remote-tracking ref, so the diff is empty, so the endpoint answered:

```
{"remote":"ghost.git","message":"Fetched from 'ghost.git': already up to date.,"updated_refs":[]}
```

`200 OK`. A fetch that reached a target it was never authorised to reach, reported as a no-op. That is the exact shape of lie ADR 0037 exists to prevent, arriving through a door ADR 0037 does not watch.

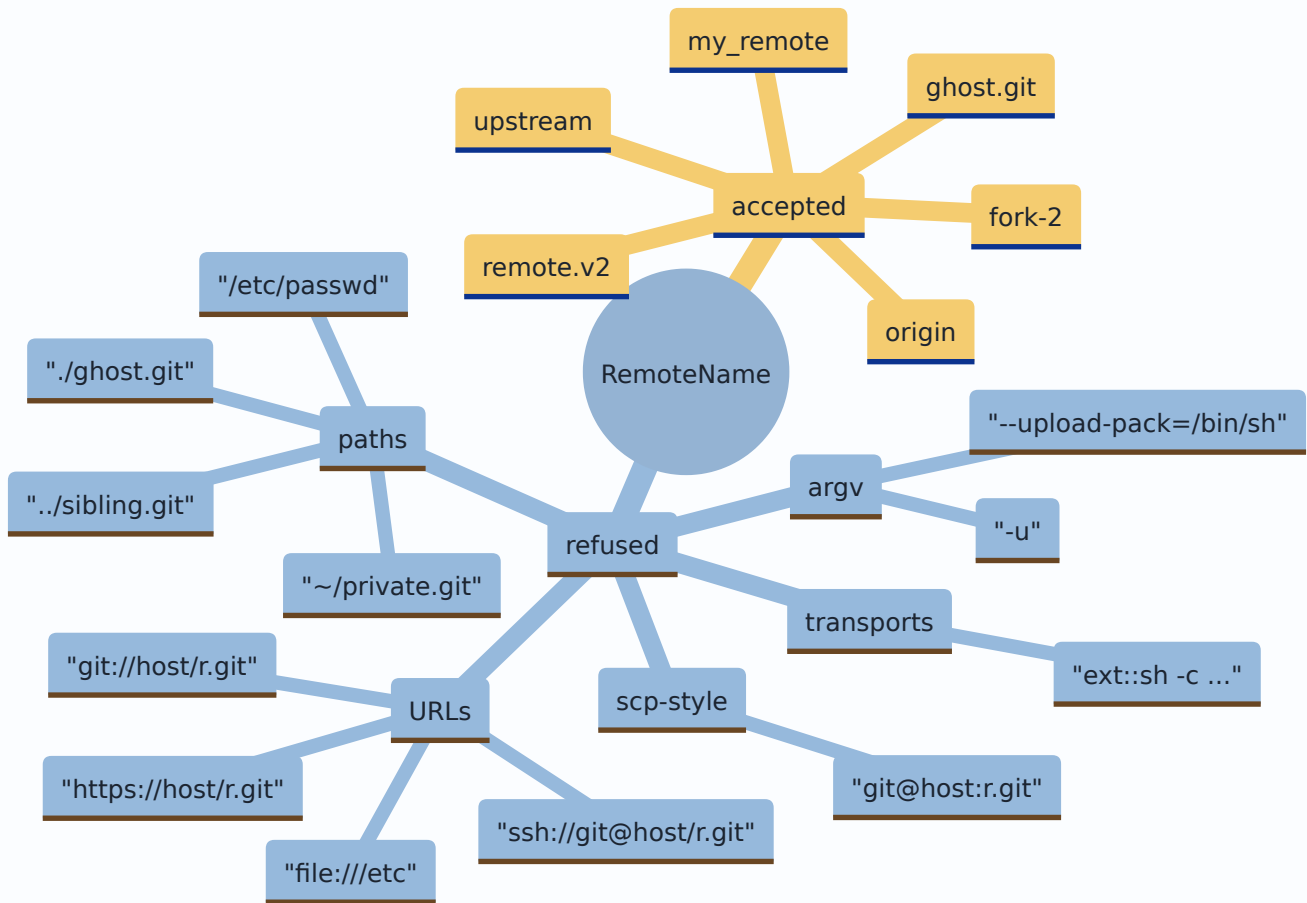
Note what does **not** help: the sandbox. ADR 0028 is explicit that the Network tier constrains *ports*, not *hosts*, and says so in terms — "this list is not an egress policy and must never be described as one". Ports 443 and 22 are open to every host on the internet. The destination boundary has to be an application-layer boundary or it does not exist.

Decision

Do both halves. They close different sets, they fail in different ways, and the cheaper one alone leaves a real hole.

(a) `RemoteName` refuses every URL and path shape

A new validator, `newtype::require_remote_name : require_git_safe`, then ≤ 100 bytes, then ASCII letters / digits / `.` / `-` / `_` only, no leading `.`, no `..` anywhere.



Every transport form needs at least one of `:`, `/`, `@`, `~` or whitespace. None of them are in the set. This is the repository's established pattern — `WorktreePath`, `CommitMessage`, `TagMessage` are all validating newtypes whose `Deserialize` runs the same validator as `new` — so it lands at the **wire boundary** and reaches every consumer of the type for free: `PullBranch`, `PushBranch`, `PushTag` and `DeleteRemoteTag` all carry a `RemoteName` field inside the `GitOperation` a submitted plan deserializes.

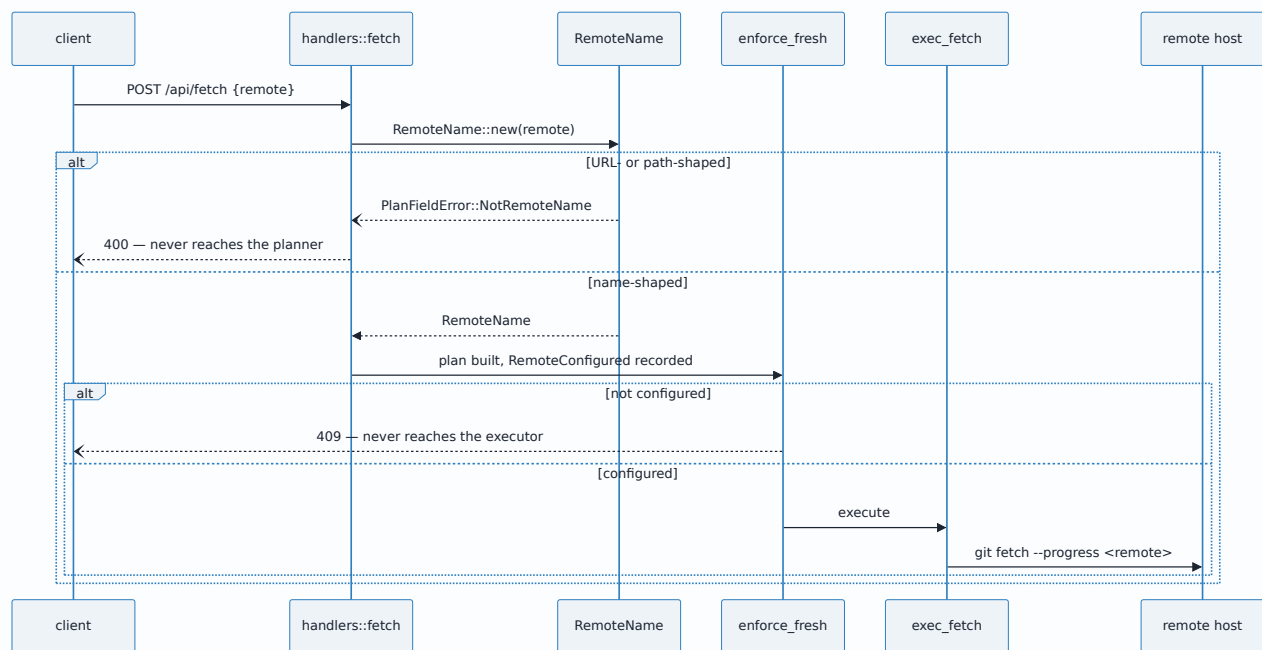
(b) A precondition with no downstream guard refuses at the gate

`planner::refuses_when_unmet_at_build(&Precondition) -> bool` — an exhaustive match with no wildcard, answering one narrow question per arm: *if this precondition is false and we run the executor anyway, does the executor refuse?*

Precondition	Executor's own guard	Verdict
RefAt / RefExists / RefAbsent	the git command refuses a missing/occupied ref	skip
BranchCheckedOut / BranchNotCheckedOut	git refuses ("cannot delete branch checked out at ...")	skip
CleanWorktree	git refuses a dirty tree	skip
SeedRecorded	exec_reset_test_repo re-reads the seed and 404s	skip
RemoteConfigured	none — git reinterprets the argument	refuse here

`enforce_fresh` gains one `else if` arm. The refusal is a `409` naming the remote:

Remote 'ghost.git' is not configured in this repository — nothing was contacted. Add it with `git remote add`, or pick a remote this repository knows.



Why each half is insufficient alone

This is the question a future reader will ask, so it is answered directly.

(a) alone leaves the unconfigured-but-well-formed case open. `ghost.git` passes every character rule any remote-name validator could reasonably impose — it is exactly the shape `git remote add ghost.git` would produce — and git resolves it as a relative path. No string rule can

consult the repository's configuration, so no amount of tightening the type reaches this. It is a smaller hazard than the URL case (a path target is inside the sandbox's filesystem grant, so it cannot reach the network) but it is the same defect: the operation ran when its stated precondition was false, and lied about the result.

(b) alone would in fact have closed the network hole — an unconfigured URL fails `RemoteConfigured` just as an unconfigured name does, and the gate refuses both. It is kept anyway, for three reasons:

1. **It is structural rather than positional.** (b) protects operations whose plan happens to carry a `RemoteConfigured` precondition. (a) protects the *type*, so a future operation, handler or internal caller that takes a `RemoteName` without that precondition cannot be pointed at a URL either. Type-level invariants do not have to be remembered at each new call site; pipeline-level ones do.
2. **It refuses earlier and more cheaply**, at deserialization, with a message that names the actual mistake rather than a downstream symptom.
3. **The two are independently verifiable**, and the mutation testing below exercises that: reverting either one alone leaves the network listener untouched, and only reverting both re-opens the connection. Two independent proofs of the same property is the belt-and-braces posture `require_git_safe`'s own doc comment already argues for.

Alternatives considered

Add an executor-side guard in `exec_fetch (git remote get-url before spawning)`. Rejected as the *primary* fix. It is a third check in a third place, it would have to be re-derived identically in `exec_pull` (#230) and `exec_push` (#231) and each tag operation, and "re-derived identically at every call site" is exactly what produced this bug — the precondition already existed and was already meant to be that check. Fixing the gate makes every current and future carrier of `RemoteConfigured` correct at once. Fixing the type makes every current and future holder of a `RemoteName` correct at once. An executor guard would have made exactly one function correct.

Make `enforce_fresh` refuse every build-time-unmet precondition. Rejected, and the existing test suite is why. `contract_suite::review_window_seed_drift_fails_closed_with_`
`the_never_recorded_refusal` asserts that a reset with no recorded seed answers 404 in the executor's own words. A blanket rule would replace that real, tested refusal — and every "fatal: a branch named 'dup' already exists" — with a paraphrase from the planner, losing git's wording for no security gain. The blocked test was information, and it pointed the right way: the classification belongs per-precondition.

Distinguish "checked and failed" from "could not be evaluated" in `held_at_build`.

Considered, and it is a genuinely good idea — it is D5's distinction (`Obs::Unknown` vs. `Obs::Absent`) applied to the precondition census, and today `held_now` flattens both into `false`. Rejected *for this*

fix because it does not close this hole: the remote here was successfully checked and genuinely was not configured, so it lands on the "checked and failed" side under either scheme and would still have been skipped. Recorded as open scope rather than done half-way.

Allow URLs but restrict them to an operator-configured allowlist. Rejected as premature: there is no allowlist mechanism, no UI for one, and no request in the product that wants an ad-hoc URL. ADR 0002 already refuses request-supplied repository paths on the same reasoning.

Consequences

A remote name is now narrower than git's own rule. `git remote add` permits anything without a slash — including spaces and `:` — so a repository with a remote named `my remote` or `weird:name` cannot be fetched through this server. Accepted deliberately: the alternative is accepting a character set that contains a transport shape, and the type's doc already recorded that in practice the only remote in play is `origin`. A user who hits this renames the remote; the refusal says exactly what the allowed set is.

Whitespace is no longer trimmed by the type. `RemoteName::new(" origin ")` is now an error, so `handlers::fetch::validate_remote`'s existing trim became load-bearing rather than cosmetic. Its doc says so.

One refusal changed layer, and its wording with it. A push or fetch naming an unconfigured remote used to reach git and return git's message; it now returns the planner's 409. Both paths are still byte-identical to each other, which is what `review_window_remote_drift_fails_closed_with_the_never_configured_refusal` actually asserts, so that test still passes — its *prose* was updated, not its assertions.

Pull inherits this verbatim, and that is why the fix lives on the fetch branch. `PullBranch` carries the same `RemoteName` and the same `RemoteConfigured` precondition, so `feature/m2.20d-pull-execution` gets both halves by rebasing rather than by re-implementing. The suite asserts it *today*, before pull's executor exists: a pull with an unconfigured remote is refused by the gate with a 409 rather than reaching pull's 501 stub — which is the assertion that will keep meaning something once the stub is replaced.

Open, recorded rather than hidden:

- **enforce_fresh's refusals are plain text, not `FetchError`.** `/api/fetch`'s typed error contract covers refusals the *handler* makes; the staleness 409 has always been a bare string, and this new refusal matches it. Making every planner refusal carry the endpoint's error type is a wider contract change than this fix.
- **held_at_build still flattens "checked and failed" into "could not check".** See the third alternative above. No current precondition is known to be affected, but the distinction is the one D5 spent a whole task establishing everywhere else.

- **The listener test's promptness window is bounded, not proved.** The watcher polls `accept()` and waits up to one second for a connection to surface before concluding none happened. A connect that somehow took longer than that would be missed. One unreproduced spurious failure was observed during development (the watcher reported a connection on a leg that never spawned git); it was not explained, and the watcher was rewritten to count connections across the call window rather than latch a bool, so a recurrence now says whether the connection arrived inside the window under test or outside it. Reported, not silently smoothed over.

What proves it

`crates/git-vista-server/src/planner/remote_boundary_suite.rs` — seven tests, all driving the real pipeline.

The load-bearing one binds a **real TCP listener** on `127.0.0.1:9418` and asserts nothing connects to it, with a **paired positive control on the same run** — the same URL, configured as a real remote, must connect. Port 9418 is not incidental: it is the only unprivileged entry in `sandbox::DEFAULT_GIT_PORTS`, so it is the only port the Network tier's Landlock connect grant covers. An ephemeral port would have asserted "nothing connected" against a connect the *sandbox* refused, and would have passed with both fixes reverted.

The precondition half is observed by effect, not by status code: `.git/FETCH_HEAD` is absent before and must be absent after, because that file is written by any fetch that reached a target. This matters because the pre-fix endpoint answered `200 OK` — a status-code assertion would have been not merely weak but wrong.

Mutation results

Every fix was reverted in turn and the suite re-run, to prove the tests bite rather than decorate.

Mutation	Caught by
Both reverted (the original hole)	6 tests — including the listener test, which reports the connection to <code>git://127.0.0.1:9418</code>
(a) only — <code>RemoteName</code> back to <code>require_git_safe</code>	<code>remote_name_refuses_every_url_and_path_shape</code> . The listener test still passes: (b) alone closes the network hole
(b) only — <code>RemoteConfigured</code> back to skip	4 tests, including <code>an_unconfigured_remote_is_never_fetched_from</code> (<code>FETCH_HEAD</code> written, endpoint answers <code>200 ... already up to date</code>) and pull's. The listener test still passes: (a) alone closes the network hole
The whole <code>else if</code> arm deleted from <code>enforce_fresh</code>	3 tests
Census over-widened (<code>SeedRecorded</code> flipped to refuse)	<code>only_remote_configured_refuses_when_unmet_at_build</code> and the rewritten planner unit test

The last row is the anti-vacuity check on the classification itself: the census pins the `true` set to exactly `{RemoteConfigured}` , so widening it is a visible edit rather than a side effect.

One existing test had to be rewritten:

`planner::tests::a_precondition_unmet_at_build_time_is_left_to_the_executor` asserted `enforce_fresh` steps aside — using a never-configured push remote as its example, and never checking that anything caught the operation afterwards. It was pinning the mechanism of this hole as though it were a guarantee. The rule it states is still the rule, so the test was kept and made stronger rather than deleted: the example moved to `RefAbsent` , a second leg now **proves** the executor's guard fires (`fatal: a branch named 'dup' already exists`), and a third leg asserts the `RemoteConfigured` exception directly.

Signed: thomas2025 · 2026-08-02T22:20:00-04:00